MAJOR PROJECT REPORT

Student Performance Prediction Using Machine Learning

Machine Learning Based Academic Performance Prediction System

Submitted By	Singipurapu Dinesh
Project Type	Major Project
Domain	Data Science / Machine Learning
Primary Algorithm	Linear Regression
Dataset Size	1,000 records × 12 columns

Prepared in accordance with the Major Project submission requirements.

Documentation includes the required project description, dataset analysis, model explanation, screenshots, results, conclusion, and future scope.

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1. Project Title

Student Performance Prediction Using Machine Learning

This project develops a machine learning system that analyzes student academic and behavioral attributes and predicts a student's final academic score. The project combines data processing, exploratory analysis, visualization, model training, evaluation, and an interactive command-line prediction workflow.

2. Problem Statement

Educational performance is influenced by multiple measurable factors such as study hours, attendance, previous academic performance, assignment performance, midterm performance, extracurricular participation, internet access, and sleep hours. Manual assessment of these relationships can be time-consuming and may not provide a consistent prediction of future performance.

The problem addressed by this project is to use historical student data and machine learning techniques to estimate a student's final score from available academic and behavioral attributes. The resulting system is intended to demonstrate how data-driven methods can support early performance analysis and academic decision-making.

3. Objective

1. Load and inspect the student performance dataset.
2. Perform data processing and understand dataset structure, data types, and completeness.
3. Conduct exploratory data analysis to identify important patterns and relationships.
4. Visualize the distribution of final scores and relationships between key variables.
5. Analyze numerical correlations to identify influential features.
6. Train a Linear Regression model using a preprocessing and regression pipeline.
7. Evaluate the model using MAE, MSE, RMSE, and R^2.
8. Generate a final-score prediction for new student information.

4. Tools & Technologies

Category	Technology / Tool
Programming Language	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Model	Linear Regression
Development Environment	Jupyter Notebook / Visual Studio Code
Dataset Format	CSV

5. Dataset Description

The project uses a CSV dataset named **student_performance.csv**. The displayed dataset contains 1,000 student records and 12 columns. The target variable is **final_score**.

Feature	Description
student_id	Unique student identifier
gender	Student gender category
age	Student age
study_hours	Study hours per day
attendance	Attendance percentage
previous_score	Previous academic score
assignment_score	Assignment performance score
midterm_score	Midterm examination score
extracurricular	Participation in extracurricular activities
internet_access	Internet access availability
sleep_hours	Sleep hours
final_score	Final academic score / prediction target

The dataset overview shown in the project confirms 1,000 non-null values for each of the 12 displayed columns, with numerical and categorical variables represented using appropriate data types.

6. Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure of the dataset, the distribution of the target variable, relationships between academic factors and final score, and correlations among numerical features.

6.1 Dataset Inspection

The dataset shape is (1000, 12), and the overview confirms the presence of academic, behavioral, and demographic variables.

6.2 Final Score Distribution

The final-score histogram shows that the majority of students fall within a central score range, while very high and very low scores occur less frequently.

6.3 Study Hours vs Final Score

The scatter plot shows a positive trend: increased study hours are generally associated with higher final scores.

6.4 Attendance vs Final Score

Higher attendance generally corresponds to better final scores, though the plot also shows variation between individual students.

6.5 Correlation Analysis

The heatmap shows that midterm score has the strongest displayed relationship with final score (0.80), followed by study hours (0.60), previous score (0.56), assignment score (0.50), and attendance (0.28).

7. Model / Algorithm Explanation

The project uses a **Linear Regression** model to predict the continuous target variable, final_score. Linear Regression estimates the target as a weighted combination of the input features. The implementation uses a Scikit-learn Pipeline so that preprocessing and model fitting can be handled consistently.

Model workflow: input dataset → feature/target separation → training/test split → preprocessing → Linear Regression training → prediction on test data → evaluation using regression metrics → prediction for new student information.

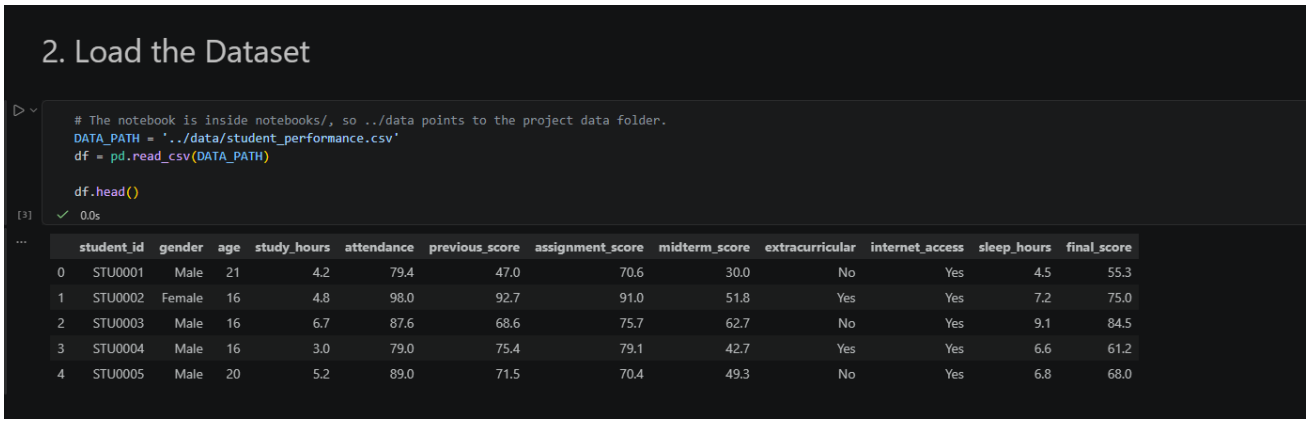
Evaluation Metrics

Metric	Obtained Value	Interpretation
MAE	2.42	Average absolute prediction error is approximately 2.42 score points.
MSE	9.54	Mean squared prediction error.
RMSE	3.09	Typical prediction error scale is approximately 3.09 score points.
R^2	0.8681	The model explains approximately 86.81% of the variance in the test-set final scores.

8. Screenshots

The following screenshots document the required stages of the project submission: data processing, exploratory visualizations, model training, model evaluation, actual-versus-predicted analysis, and final model output.

Figure 1. Dataset Loading and Sample Records



2. Load the Dataset

```
# The notebook is inside notebooks/, so ../data points to the project data folder.
DATA_PATH = '../data/student_performance.csv'
df = pd.read_csv(DATA_PATH)

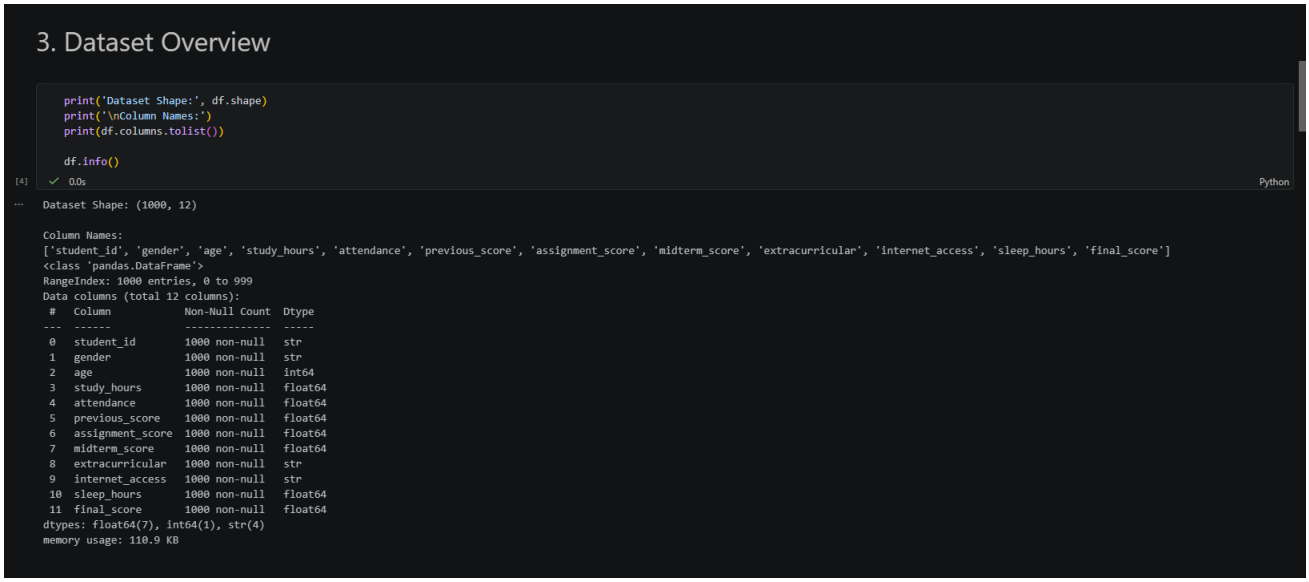
df.head()
```

[3] ✓ 0.0s

	student_id	gender	age	study_hours	attendance	previous_score	assignment_score	midterm_score	extracurricular	internet_access	sleep_hours	final_score
0	STU0001	Male	21	4.2	79.4	47.0	70.6	30.0	No	Yes	4.5	55.3
1	STU0002	Female	16	4.8	98.0	92.7	91.0	51.8	Yes	Yes	7.2	75.0
2	STU0003	Male	16	6.7	87.6	68.6	75.7	62.7	No	Yes	9.1	84.5
3	STU0004	Male	16	3.0	79.0	75.4	79.1	42.7	Yes	Yes	6.6	61.2
4	STU0005	Male	20	5.2	89.0	71.5	70.4	49.3	No	Yes	6.8	68.0

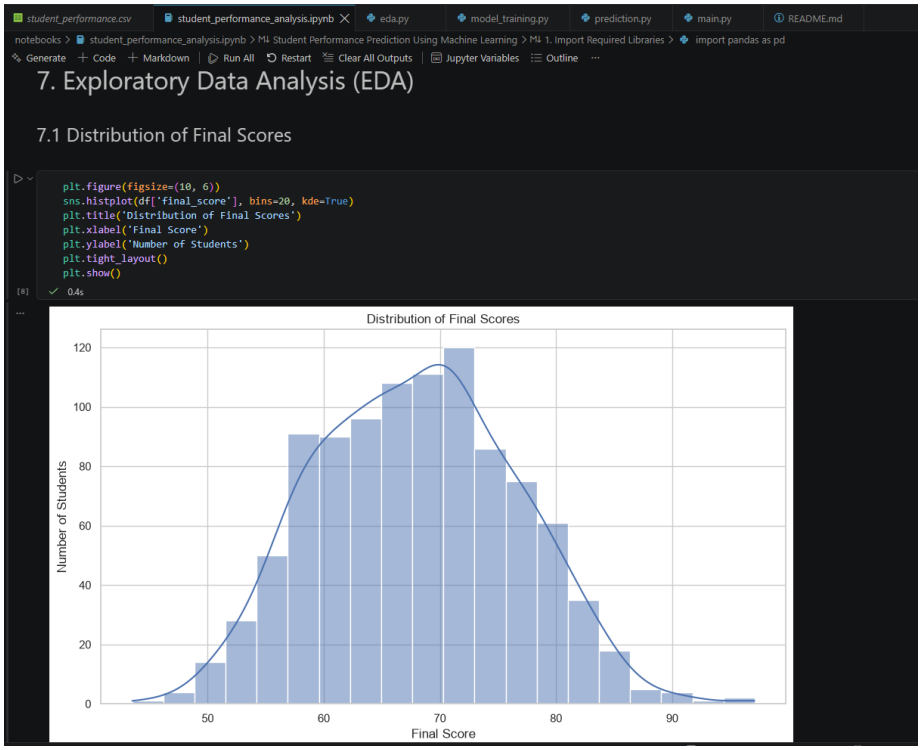
The student performance dataset is loaded from the project data folder. The displayed records show student attributes such as gender, age, study hours, attendance, previous score, assignment score, midterm score, extracurricular participation, internet access, sleep hours, and final score.

Figure 2. Dataset Overview and Data Processing



The dataset contains 1,000 records and 12 columns. The overview confirms the column names, data types, and non-null counts. All displayed fields contain 1,000 non-null observations.

Figure 3. Distribution of Final Scores



The histogram with density curve shows the distribution of students' final scores. Most observations are concentrated around the middle-to-upper score range, with fewer observations at the extremes.

Figure 4. Study Hours vs Final Score



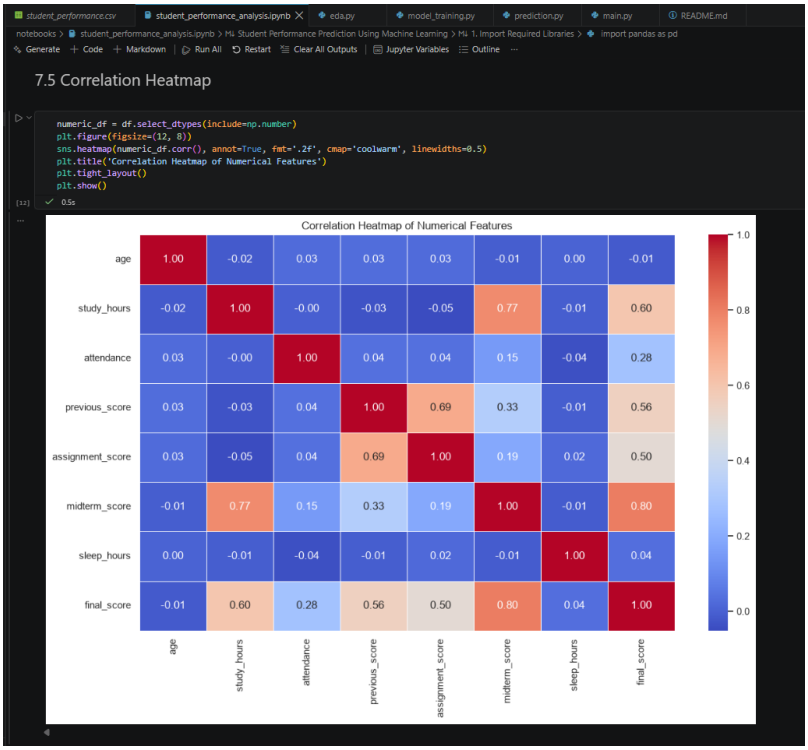
The scatter plot indicates a positive relationship between study hours and final score. Students with higher study hours generally tend to achieve higher final scores.

Figure 5. Attendance vs Final Score



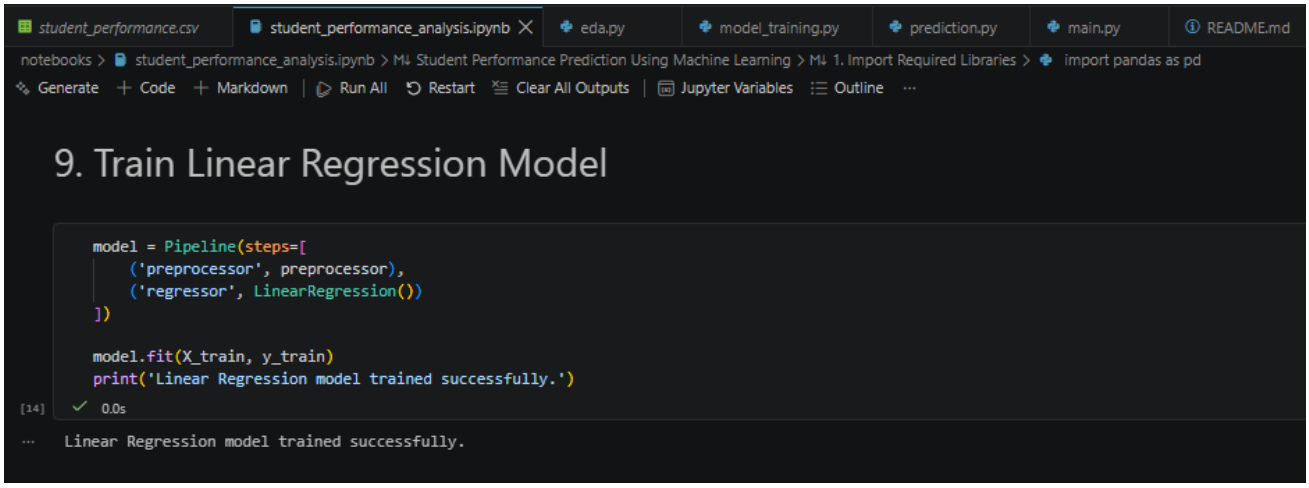
The attendance analysis shows a generally positive relationship between attendance percentage and final score, although individual performance varies considerably.

Figure 6. Correlation Heatmap



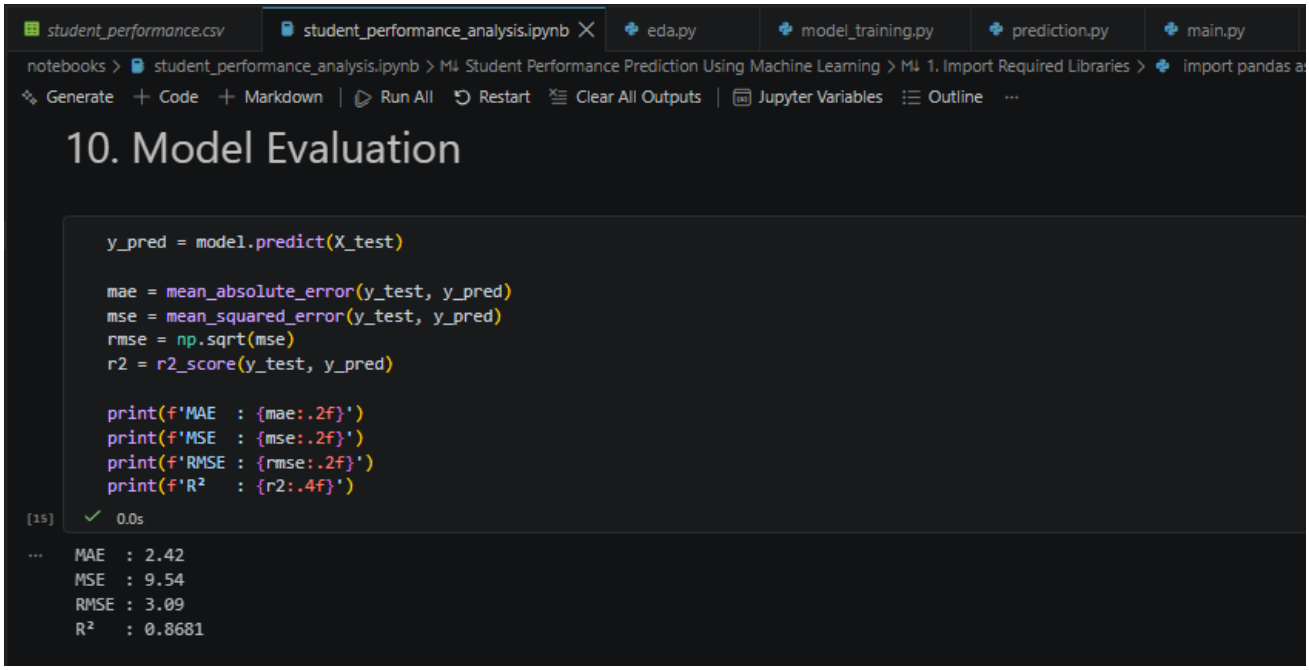
The correlation matrix highlights the relationships among numerical variables. Midterm score has the strongest displayed correlation with final score (0.80), followed by study hours (0.60), previous score (0.56), assignment score (0.50), and attendance (0.28).

Figure 7. Linear Regression Model Training



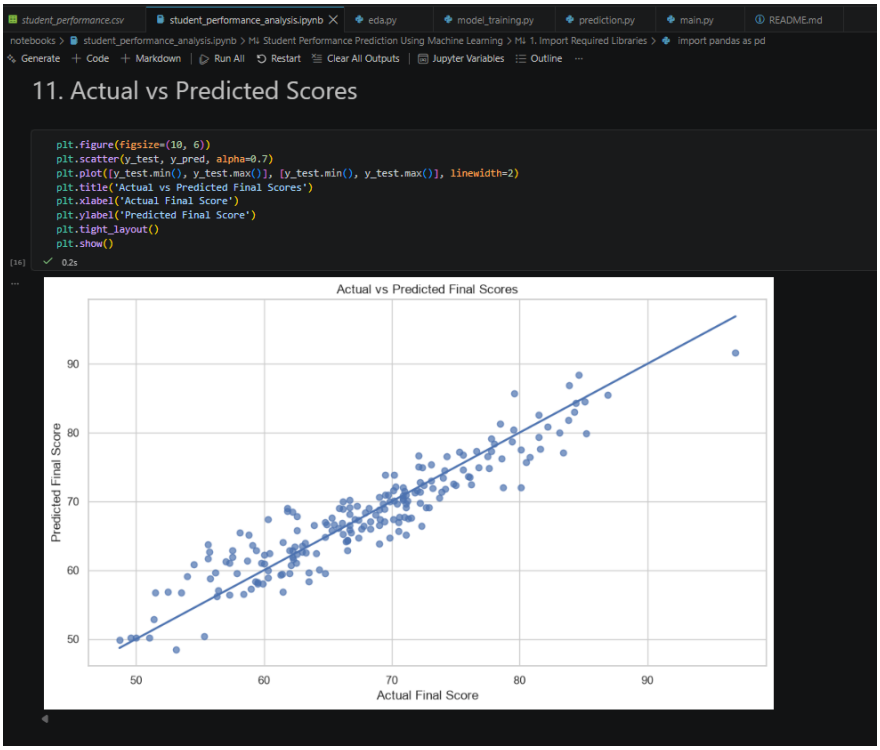
A machine learning pipeline combines preprocessing with Linear Regression. The model is trained using the training dataset.

Figure 8. Model Evaluation Results



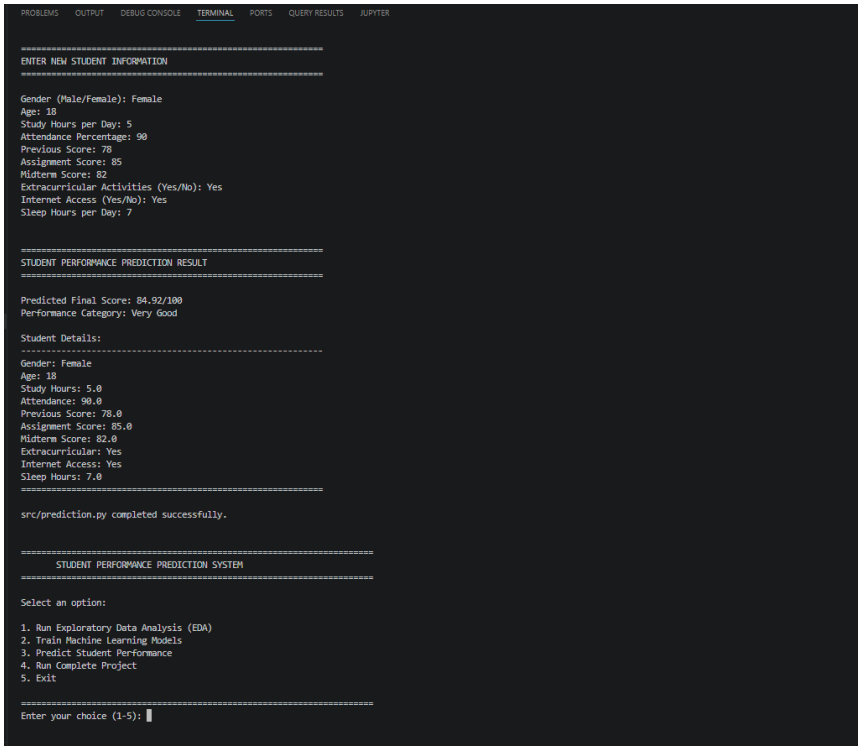
The trained model is evaluated on the test dataset. The recorded metrics are MAE = 2.42, MSE = 9.54, RMSE = 3.09, and R^2 = 0.8681.

Figure 9. Actual vs Predicted Final Scores



The actual-versus-predicted plot shows the relationship between observed final scores and model predictions. The points are concentrated around the reference line, indicating strong predictive alignment.

Figure 10. Student Performance Prediction Output



The system accepts a new student's information and produces a predicted final score. In the demonstrated run, the predicted final score is 84.92/100 and the performance category is Very Good.

9. Results & Insights

1. The project successfully loads and analyzes a 1,000-record student performance dataset.

2. The exploratory analysis identifies meaningful positive relationships between study hours and final score and between attendance and final score.

3. The correlation analysis shows that midterm score has the strongest displayed correlation with final score (0.80).

4. The Linear Regression model achieved MAE = 2.42, RMSE = 3.09, and R^2 = 0.8681 on the evaluated test data.

5. The actual-versus-predicted visualization shows predictions concentrated around the reference line, supporting the observed model performance.

6. The prediction module successfully accepts new student attributes and returns a predicted final score with a performance category.

7. In the demonstrated prediction run, the system produced a final-score prediction of 84.92/100 and categorized the performance as Very Good.

10. Conclusion

The Student Performance Prediction Using Machine Learning project demonstrates a complete data science workflow from dataset loading and exploratory analysis to machine learning model training, evaluation, and prediction. The results show that the selected features contain useful information for estimating final academic scores, and the Linear Regression model achieved an R^2 score of 0.8681 with an MAE of 2.42.

The project therefore provides a practical demonstration of how Python, data analysis, visualization, and machine learning can be combined to build a student performance prediction system.

11. Future Scope

1. Evaluate additional machine learning algorithms such as Random Forest, Gradient Boosting, and other regression models.

2. Perform systematic hyperparameter tuning and cross-validation to improve model robustness.

3. Add more student-related features where appropriate, such as historical attendance trends and additional assessment records.

4. Develop a web-based dashboard for teachers, administrators, or students.

5. Add model explainability features to show which factors contribute most to an individual prediction.

6. Store prediction history in a database for longitudinal performance monitoring.

7. Deploy the trained model as an API or cloud-based application for wider accessibility.

Submission Checklist

Requirement	Recommended File / Folder
Python code files	Project .py files (applicable source files)
Dataset	student_performance.csv
Screenshots	MajorProject_Singipurapu_Dinesh

Documentation PDF	MajorProject_Singipurapu_Dinesh.pdf